Project Plan

DormFix — Hostel Complaint & Maintenance Management System

1. Project Timeline (Original Plan)

Phase	Planned Duration	Planned Deliverables	Final Status
Phase 1: Requirements & Design	Week 1	Problem statement, SRS, use cases, architecture, and database design	Completed
Phase 2: Core Backend Setup	Week 1	Project scaffolding, registration, login, JWT authentication, and role-based middleware	Completed
Phase 3: Complaint Feature	Week 2	Complaint submission, resident dashboard, and image upload	Completed
Phase 4: Warden Features	Week 2	Warden dashboard, filtering, sorting, manual reordering, and status updates	Completed
Phase 5: Feedback & UI Polish	Week 3	Feedback system, interface refinement, and error handling	Completed
Phase 6: Testing	Week 4	Test case execution, validation, and bug fixing	Completed
Phase 7: Documentation & Demo	Week 5	Project report, demo video, and final packaging	Completed
Phase 8: Buffer / Contingency	Week 5	Review and resolution of remaining issues	Completed

2. Final Progress Against Plan

The team has completed all planned project phases, covering requirements analysis, system design, backend and frontend development, testing, documentation, and demonstration preparation.

The completed deliverables include:

- **Requirements and design:** Problem statement, Software Requirements Specification (SRS), use cases, architecture, database design, and UML diagrams.
- **Authentication and authorization:** Resident registration and login, password hashing, JWT-based authentication, protected routes, and role-based access control.
- **Complaint management:** Complaint submission, optional image attachments, complaint history, status tracking, and complaint ownership validation.
- **Warden functionality:** Dashboard statistics, complaint filtering and sorting, manual prioritization, status management, and daily resolution trends.

- **Feedback management:** Ratings from 1 to 5, optional comments for resolved complaints, and duplicate-feedback prevention.
- **Testing and validation:** Test case preparation, execution, defect identification, and verification of the implemented workflows.
- **Documentation and demonstration:** Project report, project management records, contribution details, Git evidence, screenshots, and demo video preparation.
- **Final review:** Review of the application, documentation, and submission package.

The project has reached completion of its planned scope, with the implemented system supporting the principal workflows for both residents and wardens.

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3. Risks and Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Team members unavailable near the final deadline	Medium	High	Distributed responsibilities and coordinated development helped complete the planned work.
MongoDB connection or environment issues during demonstration	Low	Medium	Database configuration and connection settings were reviewed, and the application environment was checked.
Scope creep from additional features	Medium	Medium	The documented project scope was followed, and out-of-scope features were excluded from the implementation.
Merge conflicts during parallel frontend and backend development	Low	Medium	Modular development and defined responsibilities helped reduce conflicts between components.
Invalid access to another resident's complaints	Medium	High	Authentication, role-based authorization, and complaint ownership checks were incorporated into the application.
Invalid complaint status transitions or duplicate feedback	Medium	Medium	Server-side validation and eligibility checks were implemented to enforce workflow rules.

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4. Task Allocation Summary

The project responsibilities were distributed among the four team members according to their assigned implementation and documentation work.

Area	Primary Owner	Responsibilities
Backend: Authentication and Complaint Services	Prakriti Gupta	Authentication implementation, complaint backend, and SRS preparation

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Authorization and Frontend Interfaces	Paramita Ghosh	Role-based authorization, resident complaint interface, warden interface, project report, and project management documentation
Feedback and Testing	Naina	Feedback-related work, test case preparation, and testing activities
UML and System Modelling	Nishi Gupta	UML diagrams and supporting system design documentation

6. Project Completion Summary

DormFix has completed the planned requirements, design, implementation, testing, and documentation activities. The system provides a centralized workflow for residents to register and track hostel complaints and for wardens to manage complaint status, prioritize work, review dashboard statistics, and receive feedback on resolved complaints.

The project follows a modular client-server architecture using the MERN stack, with authentication, authorization, server-side validation, and persistent database storage supporting the main workflows.

The original project plan provided a structured basis for task allocation and progress monitoring. Completion of the planned phases represents the delivery of the intended project scope.